PES UNIVERSITY

Department of Computer Science & Engineering

Object Oriented Analysis and Design

UE23CS352B

Assignment 1 Submission

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Object Oriented Analysis and Design (UE23CS352B) - Lab 1

Case Study on Use Case Diagram

Case Study 1: Library Management System

Actors:

1. Member
2. Librarian
3. Administrator

Use Cases

Member

1. Search books
2. Borrow books
3. Return books

Librarian

1. Manage books
2. Monitor transactions
3. Calculate fine

Administrator

1. User management
2. System operations

Common for all actors

1. Register
2. Login

Searching

1. Search by title
2. Search by author
3. Search by category
4. Check book availability

Books Management

1. Add books
2. Update books
3. Remove books
4. Maintain member records

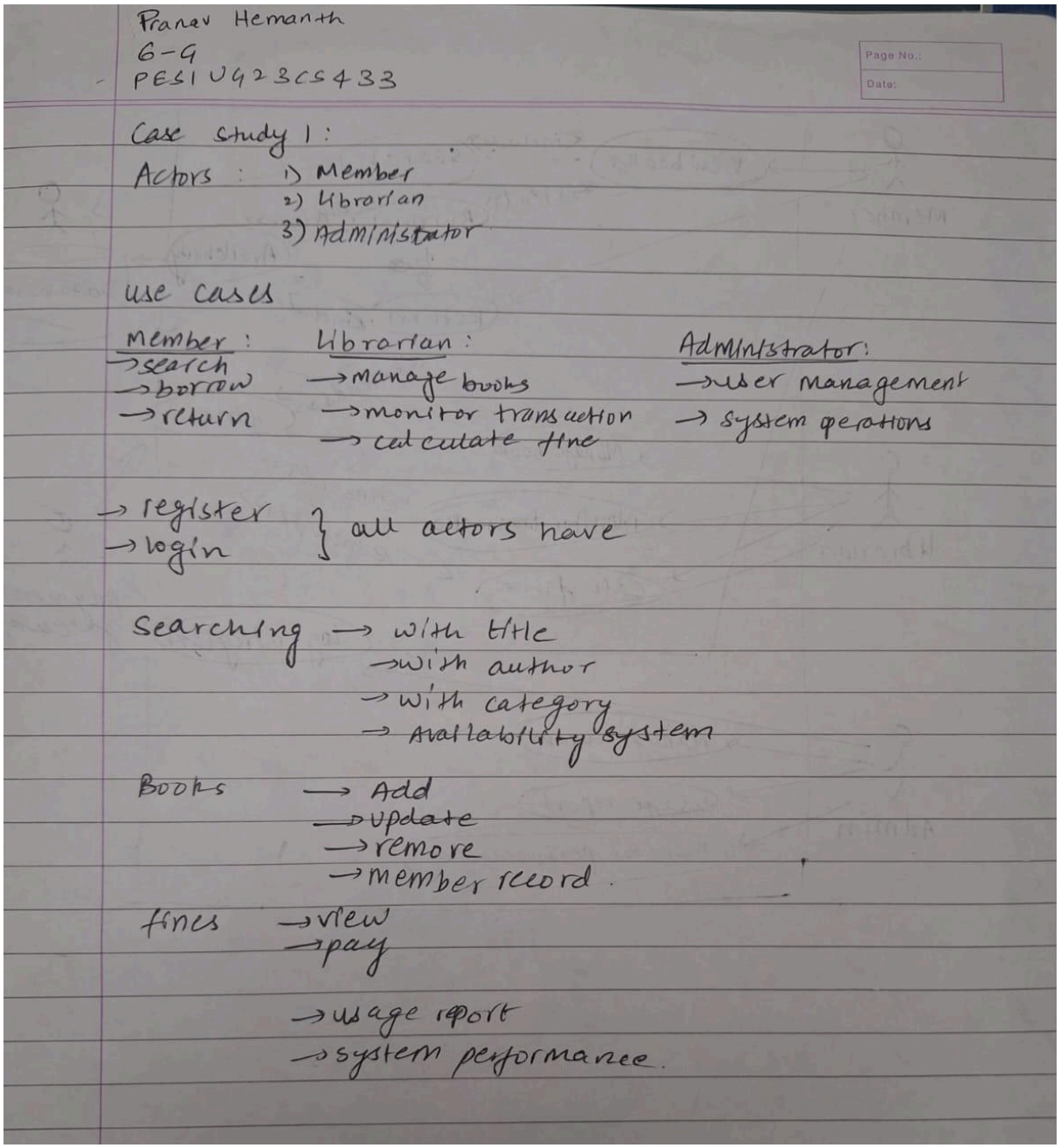
Fines

1. View fine
2. Pay fine

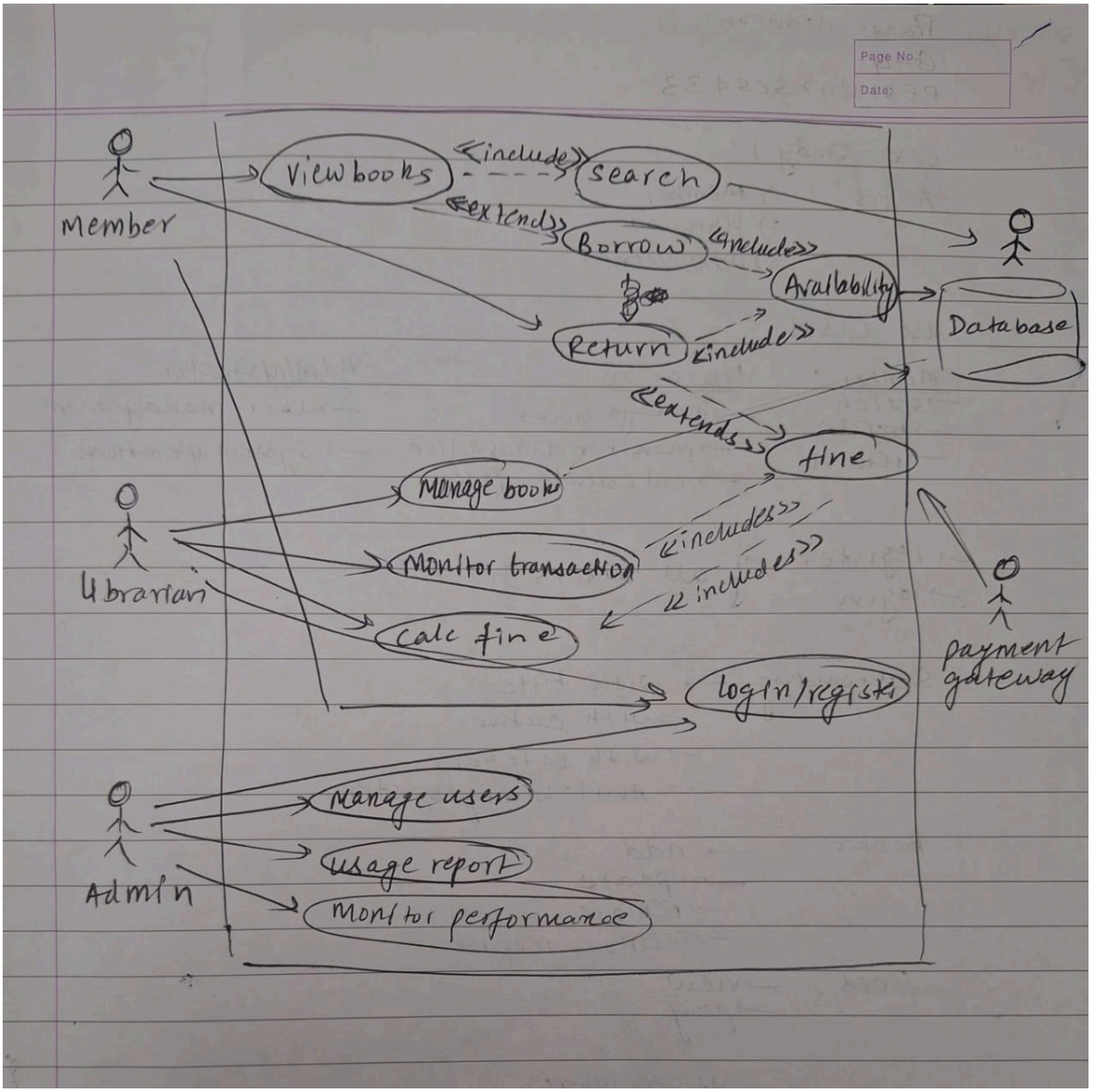
Reports and Monitoring

1. Usage report
2. System performance

Same outlined in class:



Use Case diagram:



Use Case 1

Name: Search Book
Summary: The member searches for books in the library using different search criteria and views their availability.
Actors: Member, Database
Preconditions: The member must be registered and logged in. The library catalog must exist in the system.
Description:
The member enters a search query.
The system allows searching by title, author, or category.

The system retrieves matching books from the database.
The system displays the list of books along with their availability status.
Exception:
Power Failure: If power fails while searching, the search results are not displayed.
System Failure: If the database cannot be accessed, the user is informed to try again later.
Alternative Flow:
No Results Found: If no matching books are found, the system displays a message indicating no books are available.
Post Conditions:
The list of matching books and their availability is displayed to the member.

Use Case 2

Name: Manage Books
Summary: The librarian manages the library's book collection by adding, updating, or removing books from the system.
Actors: Librarian, Database
Preconditions:
The librarian must be registered and logged in.
The librarian must have permission to manage books.
Description:
The librarian selects the manage books option.
The librarian chooses to add, update, or remove a book.
The librarian enters or modifies the book details.
The system validates the entered data.
The system updates the book records in the database.
Exception:
Power Failure: If power fails during data update, the changes are not saved.
System Failure: If the database is not accessible, the librarian is notified.
Alternative Flow:
Duplicate Book: If the book already exists, the system rejects the entry.
Invalid Data: If the entered details are incorrect, the system asks the librarian to re enter the data.
Post Conditions:
The book records in the system are updated successfully.

Case Study 2: Online Hospital Appointment & Care Management

Actors:

1. Doctors
2. Patients
3. Staff
4. Administrators
5. System (Notification Service)

Use Cases

Patients

1. Login / Register
2. Search doctors
3. Book appointment
4. Reschedule appointment
5. Cancel appointment
6. View appointment history
7. View prescriptions
8. Download reports
9. Raise queries or issues

Doctors

1. Manage schedule
2. View appointments
3. Access patient history
4. Update diagnosis
5. Prescribe medication

Staff

1. Upload test results

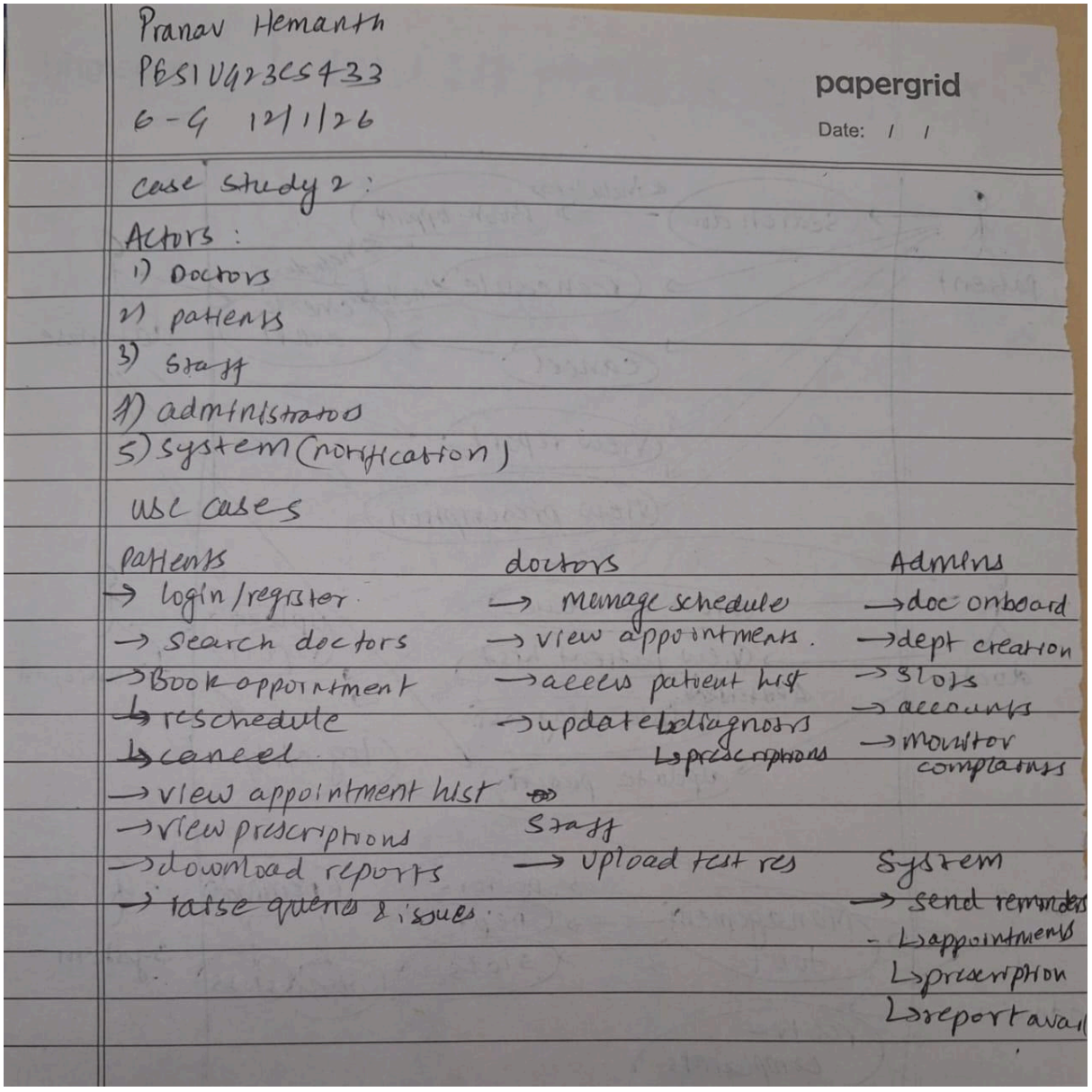
Administrators

1. Doctor onboarding
2. Department creation
3. Slot creation
4. Account management
5. Monitor complaints

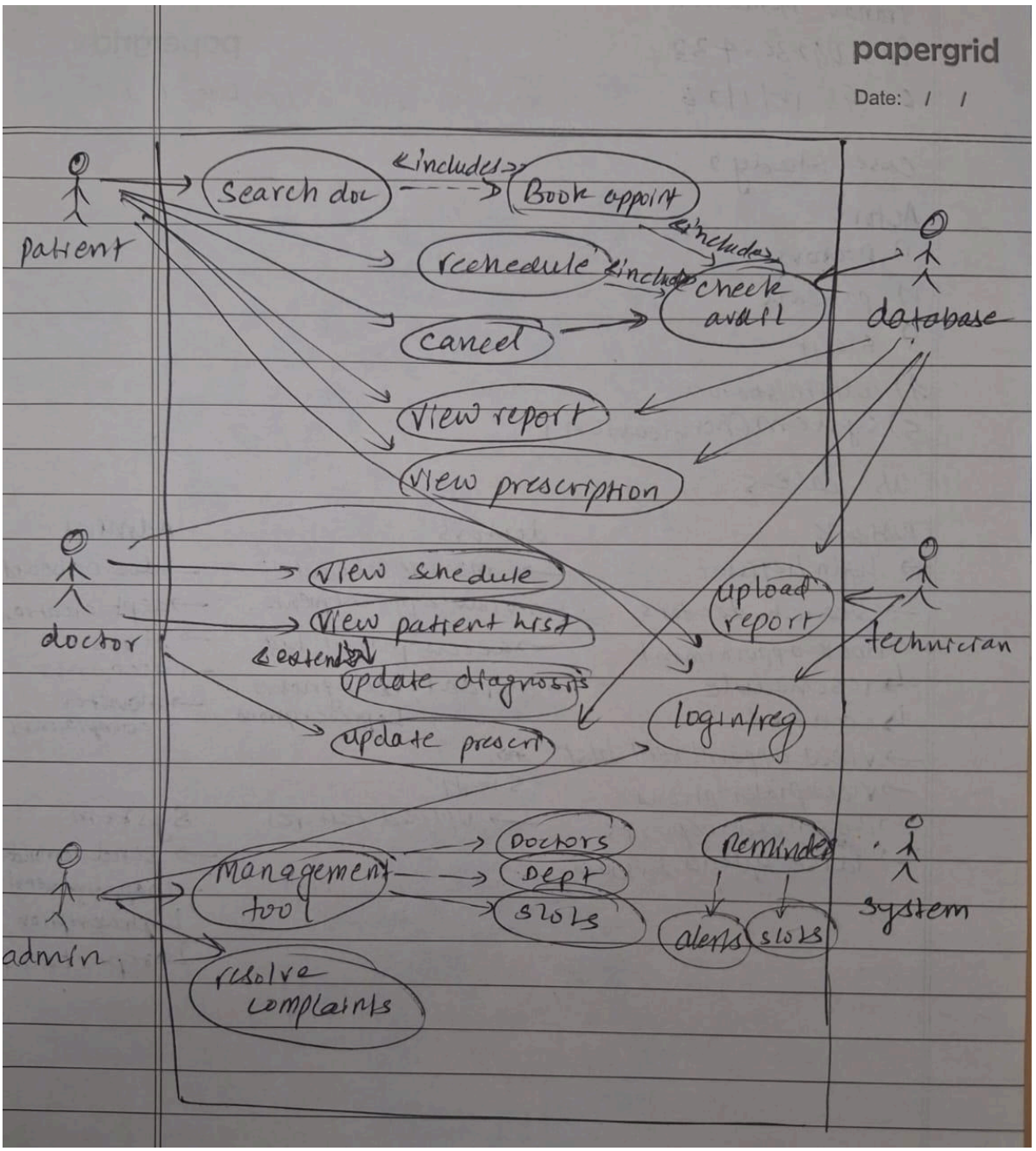
System (Notification Service)

1. Send appointment reminders
2. Send prescription alerts
3. Send report availability notifications

Same outlined in class:



Use Case diagram:



Use Case 1

Name: Book Appointment
Summary: The patient selects a doctor and a preferred time slot. The system checks the doctor's availability and, if the slot is free, the appointment is booked and stored in the system.
Actors: Patient, Doctor, Database, System (Notification)
Preconditions:
The patient must be registered and logged in.
The doctor must be registered in the system.

The selected time slot must exist.
Description:
The patient searches for a doctor.
The system displays the list of available doctors.
The patient selects a doctor and a preferred time slot.
The system verifies the doctor's availability.
The system stores the appointment details in the database.
The system sends an appointment confirmation to the patient and doctor.
Exception:
Power Failure: If power fails during booking, the appointment is not saved.
System Failure: If the system cannot access the database, the patient is asked to try again later.
Alternative Flow:
Slot Unavailable: If the selected time slot is not available, the system displays alternate slots.
Post Conditions:
The appointment is successfully recorded.
The patient and doctor are notified.

Use Case 2

Name: Update Diagnosis and Prescription
Summary: The doctor records the diagnosis and prescribes medication after examining the patient.
Actors: Doctor, Patient, Database, System (Notification)
Preconditions:
The doctor must be logged in.
The patient must have a valid appointment.
Description:
The doctor accesses the patient's medical record.
The system retrieves the patient's medical history.
The doctor enters the diagnosis details.
The doctor adds prescription information.
The system saves the data in the database.
The system notifies the patient.
Exception:
Power Failure: If power fails while saving data, the diagnosis and prescription are not stored.
System Failure: If the system cannot access the database, the doctor is notified.
Alternative Flow:
Edit Prescription: The doctor may modify the prescription before saving.
Post Conditions:
The diagnosis and prescription are stored in the system.
The patient can view the updated prescription online.